(Please write your enrollment number immediately)

Mid-Term Examination

B.Tech Programmes (Under the Aegis of USICT)
6th Semester, (March 2025)

Paper Code: CIE 310T **Subject: Advance DBMS**

Time: **1.5 Hours** Max Marks: **30**

Note: Attempt Q.1, which is compulsory and any two more questions from the remaining.

Q.No	Question	Max Marks	CO
Q1	a) Give any 2 significant impacts of emerging database standards on distributed database management systems (DDBMS). b) Discuss the role of a Storage Manager in a DBMS. c) Why are key constraints important in a distributed database management system? d) Explain any one query optimization challenge in a distributed database. e) What is a database trigger and state any one common use case for triggers?	2*5 =10	CO1 CO1 CO2 CO2 CO1
Q2	a) Differentiate between Embedded SQL and Dynamic SQL with examples. b) Explain the concept of Data Replication and its significance in distributed databases.	5 5	CO1 CO2
Q3	a)How does Parallel Database Architecture improve query performance? Explain with a diagram. b) A company needs to ensure secure database transactions. Propose a solution that involves security considerations and integrity constraints.	5 5	CO1 CO2
Q4	a) What are ACID properties in database transaction processing? Explain their importance with an example. b) Consider a relational database with **Orders (Order_ID, Customer_ID, Amount)** and **Customers (Customer_ID, Name, Address)**. Make sample tables on your own. **i.** Write an SQL query to retrieve the names of customers who have placed orders of more than $5000. **ii.** Count the number of orders placed by each customer, displaying Customer Name and Order Count.	5 5	CO1 CO2